# Probability Distribution

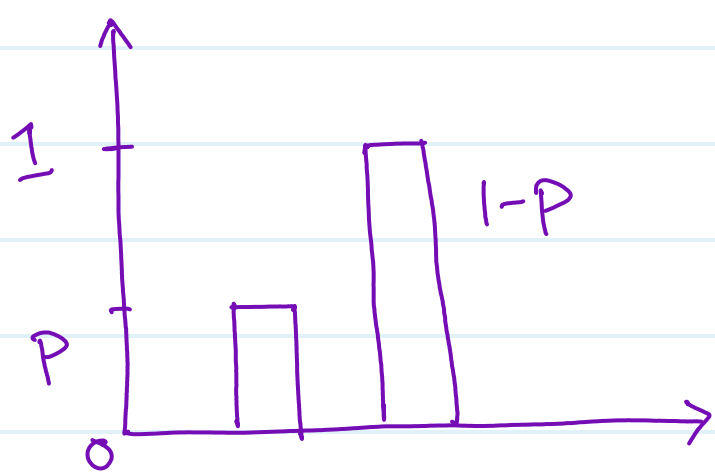
① Descret prob. Dist.

- I Bernaulli's prob. dist.
- II Binomial prob. dist.
- III Poission prob. dist.

## I) Bernaulli prob. dist. –

no. trials = single

outcome = two (fix)



II. Binomial prob. dist.

no. trials = more than one (fix)

outcome = two (fix)

III. poission prob. dist. →

$\lambda = 5$

$K = 3$

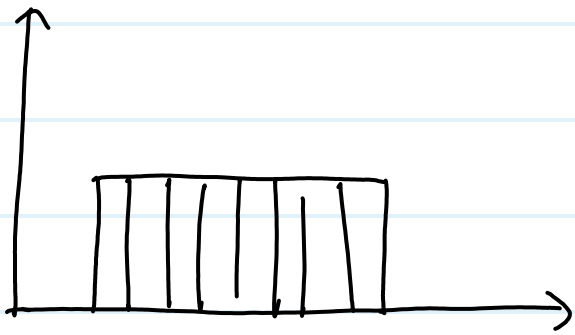
$$p(K=3) = \frac{e^{-\lambda} \lambda^{K}}{K!}$$

$$= \frac{e^{-5} 5^{3}}{3 \times 2 \times 1}$$

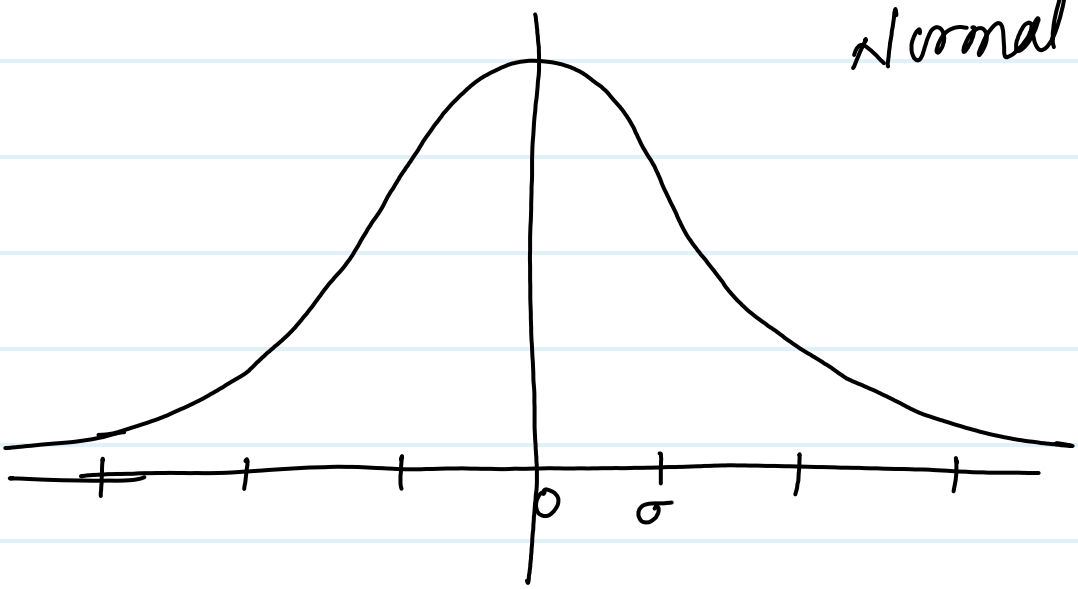
$\Rightarrow 14\%$

## 2) Continous prob dist.

### I.) uniform dist. –

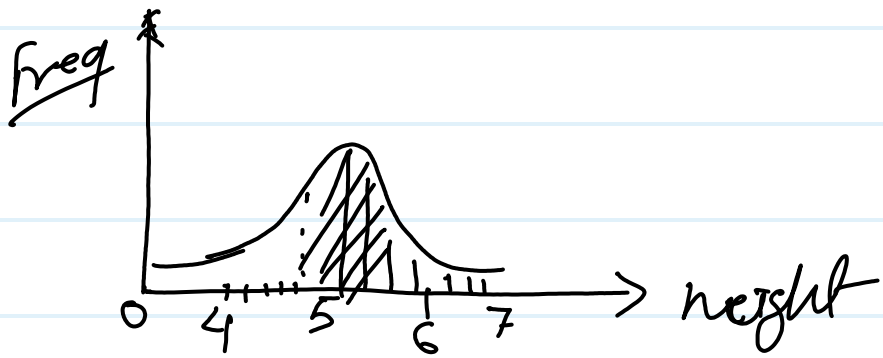


### II) Gaussian dist. – Bell curv / Normal Dist.

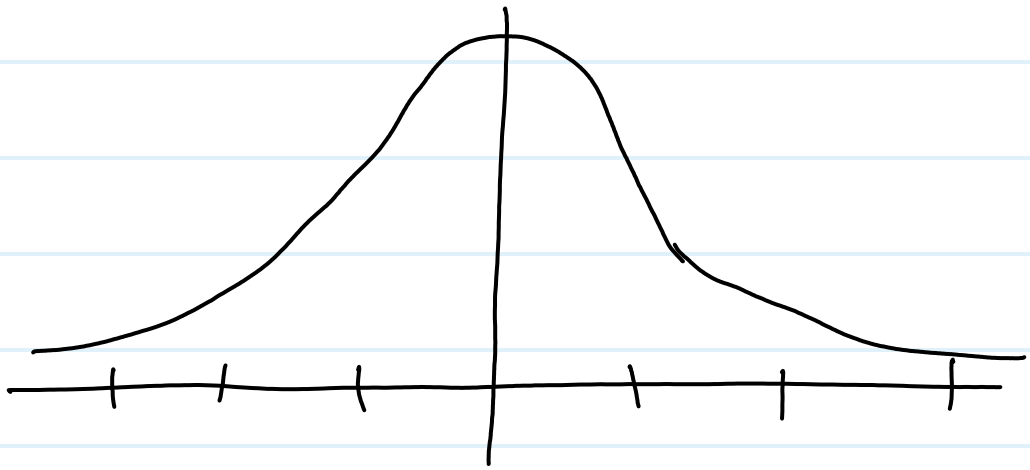



$m = m = m$

a)



## III. Standard normal dist. –



mean = 0

SD = 1

# $Z_{score} = \frac{x_i - \mu}{\sigma}$

# Feature scalling technique

| Age | Salary | height |
|---|---|---|
| 10 | 10000 | 150 |
| 20 | 20000 | 155.6 |
| 35 | 70000 | 162.3 |
| 40 | 100000 | 170 |
| 50 | 500000 | 180 |

$-3 \quad 0 \quad +3$

<u>Dataset</u> = $[1\ 2\ 3\ 4\ 5]$

$\bar{x} = 3$

$$\frac{x_i - \mu}{\sigma}$$

$x_i - \bar{x}$

$1 - 3 = -2$

$2 - 3 = -1$

$3 - 3 = 0$

$4 - 3 = 1$

$5 - 3 = 2$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$= \frac{(-2)^2 + (-1)^2 + 0 + (1)^2 + (2)^2}{5-1}$$

$$\sigma = \sqrt{\frac{4+1+1+4}{4}} \Rightarrow$$

$$= \sqrt{2.5} \Rightarrow 1.58$$

$-3$

$$\Rightarrow \frac{-2}{1.58} = -1.26$$

$$\Rightarrow \frac{-1}{1.58} = -0.63$$

$$\Rightarrow \frac{0}{1.58} = 0$$

$$\Rightarrow \frac{1}{1.58} = 0.63$$

$$\Rightarrow \frac{2}{1.58} = 1.26$$

$+3$

① Standarization (Zscore)

② Normalization (min-max)

0 - 1

max = 500000

$$= \frac{20}{500000} \Rightarrow 0.00004$$

$$= \frac{80}{500000} = 0.000016$$

$$= \frac{200000}{500000} = 0.4$$

$$= \frac{170}{500000} = 0.00034$$

$$= \frac{500000}{500000} = 1$$

$$\Rightarrow \frac{x_i - min}{max - min}$$

| Age |
|---|
| 10 |
| 20 |
| 40 |
| 60 |
| 70 |
| 50 |

min = 10

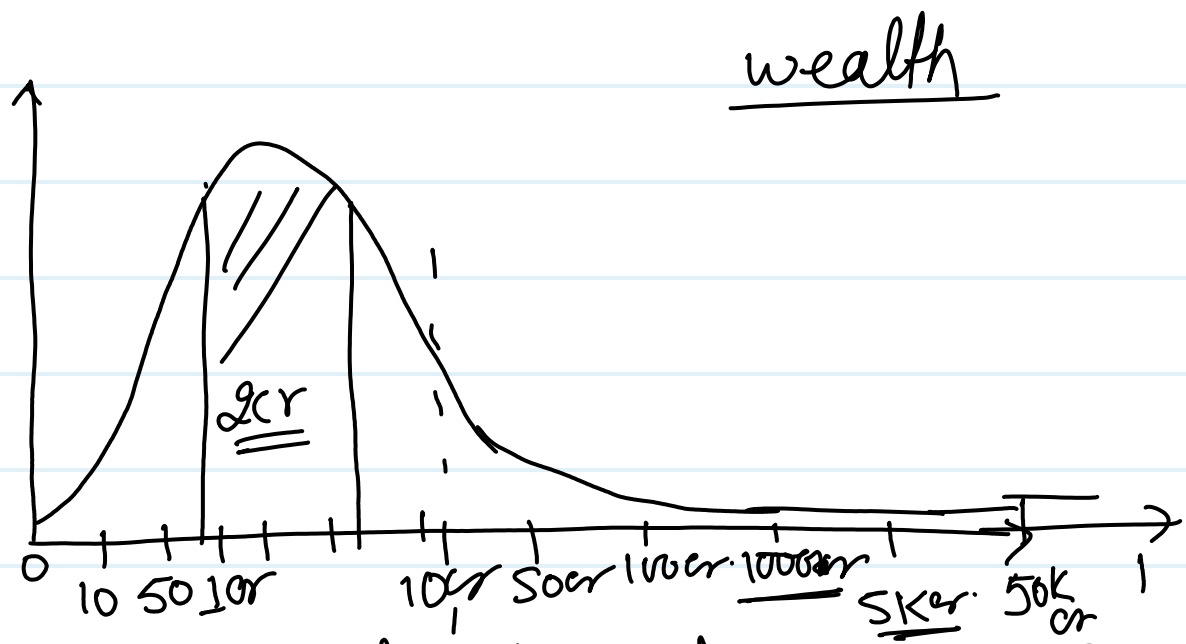
max = 70

$$\frac{10-10}{70-10} \Rightarrow 0$$

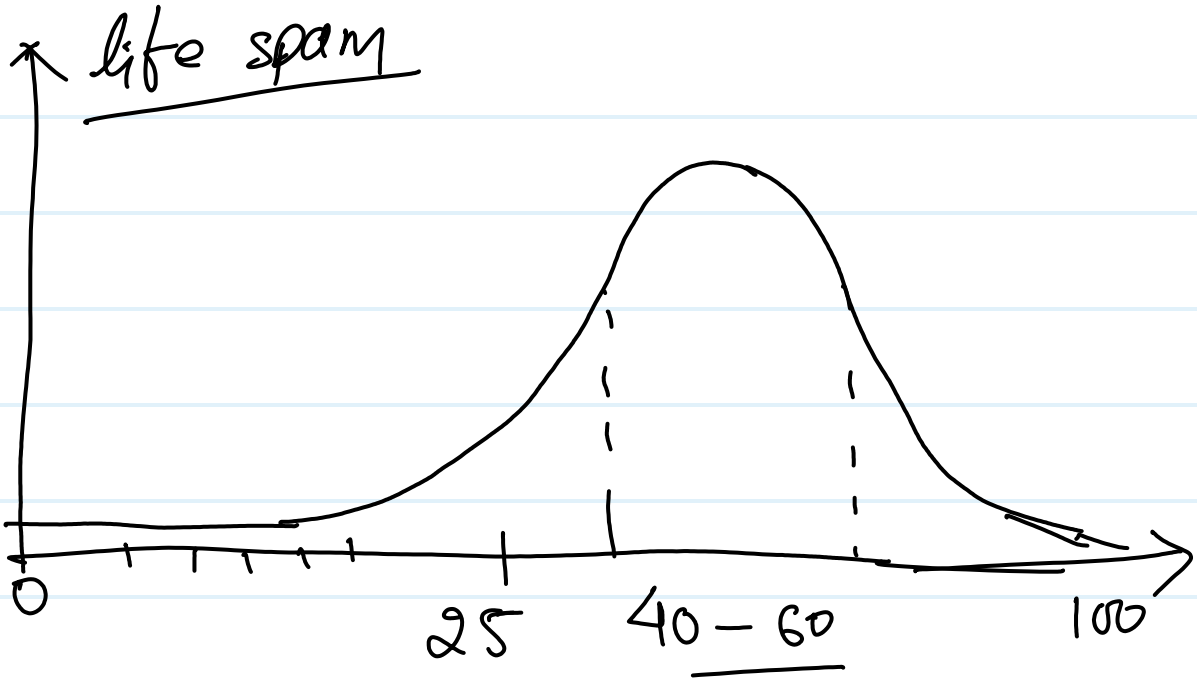
$$\frac{20-10}{70-10} = \frac{10}{60} \Rightarrow 0.16$$

$$\frac{70-10}{70-10} = \frac{60}{60} \Rightarrow 1$$

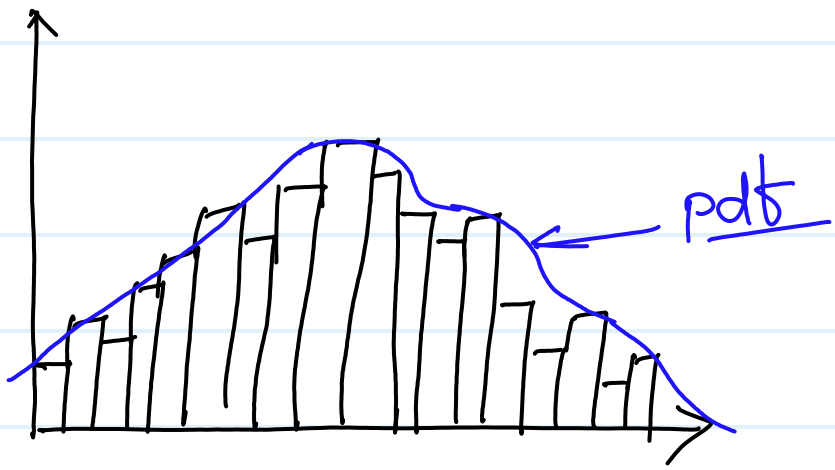
IV. log normal dist. –



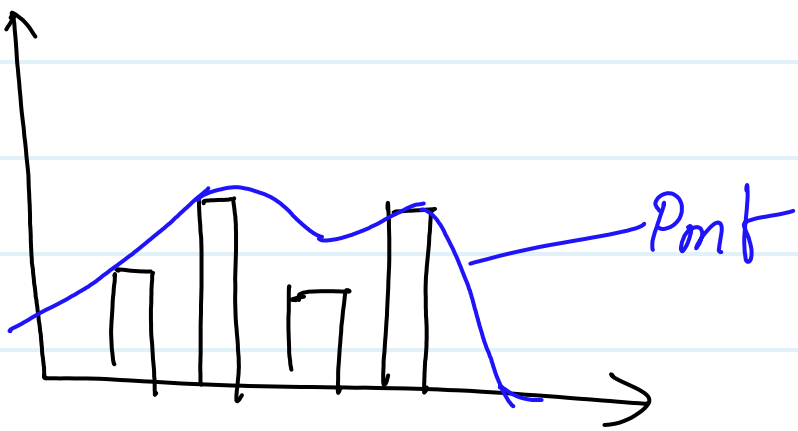

mean > median > mode


life span
0
25
40 – 60
100

# pdf probability density function




# Pmf Probability mass function




# cdf cummulative density function

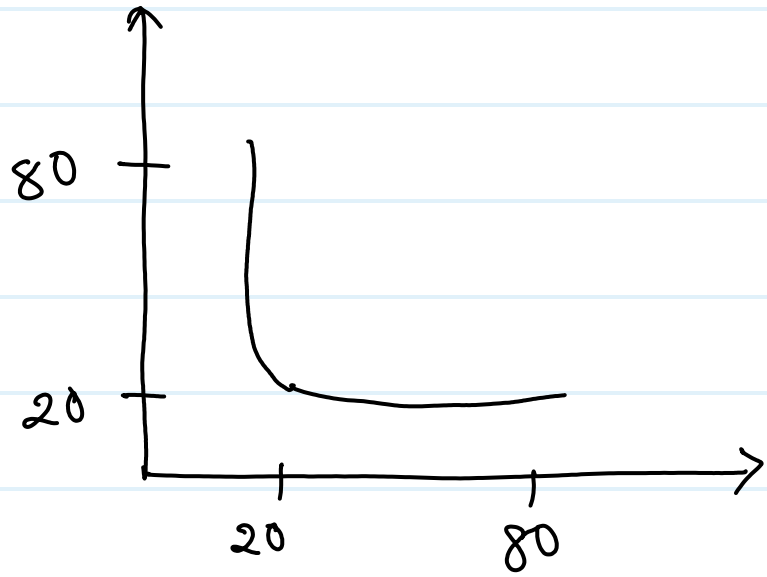
| | |
|---|---|
| 1 | 1 |
| 2 | 1+2 = 3 |
| 3 | 3+3 = 6 |
| 0 | 6+0 = 6 |
| 1 | 6+1 = 7 |
| 2 | 7+2 = 9 |
| 0 | 9+0 = 9 |

# Power law / Pareto distribution
(80-20)

family income = 80% 20%
=> 20% 80%

cricket => 80% 20%
20% 80%

workplace => 80% 20%
=> 20% 80%
↑ ↑




① Box-cox transformati
② log transformation